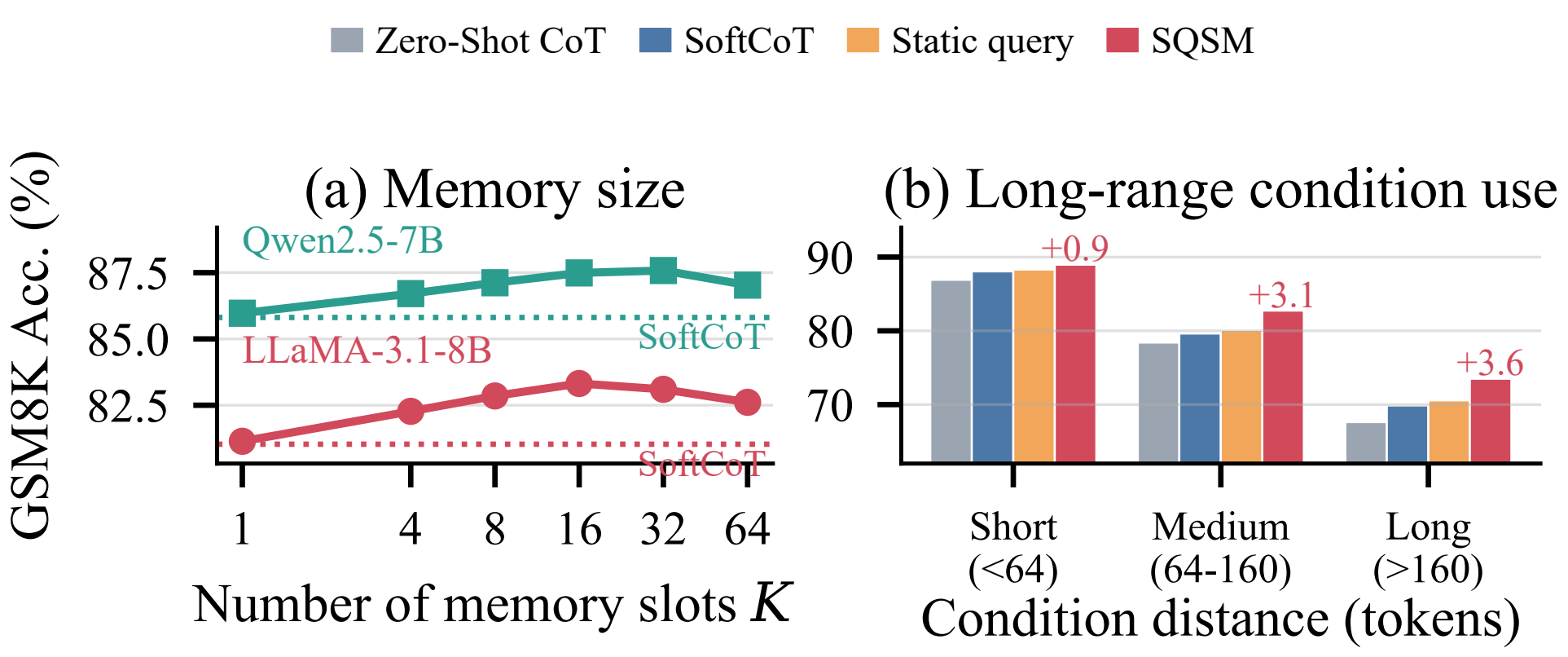
Zero-Shot CoT
SoftCoT
Static query
SQSM
GSM8K Acc. (%)
(a) Memory size
Qwen2.5-7B
LLaMA-3.1-8B
SoftCoT
SoftCoT
87.5
85.0
82.5
1
4
8
16
32
64
Number of memory slots K
(b) Long-range condition use
90
80
70
+0.9
+3.1
+3.6
Short
(<64)
Medium
(64-160)
Long
(>160)
Condition distance (tokens)